

Complex Analysis — Rule-Compliant Slice

KB pipeline · vertical slice

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1. Regions in the Complex Plane

★ ◆ DEFINITION 1.1

A set S is **connected** if it cannot be expressed as the union of two disjoint nonempty open sets.

★ ◆ DEFINITION 1.2

A **domain** is a nonempty open connected set.

★ ◆ PROPOSITION 1.3

Assume that D is a domain and $z_1, z_2 \in D$. Then there is a polygonal path from z_1 to z_2 lying in D .

2. Harmonic Functions

★ ◆ DEFINITION 2.1

Let D be a domain. A function $u(x, y)$ is **harmonic** on D if it has continuous partial derivatives of the first and second order in D and satisfies Laplace's equation in D .

★ ◆ DEFINITION 2.2

Laplace's equation is the equation $\nabla^2 u = 0$.

★ ◆ THEOREM 2.3

Assume that $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D . Then u and v are harmonic in D .

Proof. Assume that $f(z) = u(x, y) + iv(x, y)$ is analytic in a domain D . Then u and v have continuous partial derivatives of all orders in D . Then $u_x = v_y$ and $u_y = -v_x$ in D . Then $u_{xx} = v_{yx}$ and $u_{yy} = -v_{xy}$, by direct computation. Then $v_{xy} = v_{yx}$. Then u is harmonic in D . Then v is harmonic in D . Hence u and v are harmonic in D . ■

★ ◆ DEFINITION 2.4

Let u and v be functions on a domain D . The function v is **harmonic conjugate** of u in D if u and v are harmonic in D and their first-order partial derivatives satisfy the Cauchy–Riemann equations $u_x = v_y$ and $u_y = -v_x$.

★ ◆ THEOREM 2.5

Assume that $f(z) = u(x, y) + iv(x, y)$ is defined on a domain D . Then f is analytic in D if and only if v is a harmonic conjugate of u in D .

Proof. Assume that $f(z) = u(x, y) + iv(x, y)$ is defined on a domain D . ($\Rightarrow$) Therefore u and v are harmonic in D , by the fact that the real and imaginary parts of an analytic function are harmonic in its domain. Then the Cauchy–Riemann equations hold in D . Therefore v is a harmonic conjugate of u in D . ($\Leftarrow$) Then f is analytic in D . Hence f is analytic in D if and only if v is a harmonic conjugate of u in D . ■

★ ▮ CLAIM 2.6

Being a harmonic conjugate is not a symmetric relation: there exist u, v such that v is a harmonic conjugate of u but u is not a harmonic conjugate of v .

Proof. Construct the pair $u = x^2 - y^2$, $v = 2xy$ explicitly. Then v is a harmonic conjugate of u in $\mathbb{C}$. Then $g = 2xy + i(x^2 - y^2)$ fails the Cauchy–Riemann equations off the origin, by direct computation. Therefore g is not analytic in any domain, by the fact that a function is analytic in a domain if and only if its imaginary part is a harmonic conjugate of its real part there. Hence the relation is not symmetric. ■

★ ◇ PROPOSITION 2.7

Assume that u is harmonic in a simply connected domain D . Then u has a harmonic conjugate in D .

★ ▮ CLAIM 2.8

There exist a domain D and a function u harmonic in D such that u has no harmonic conjugate in D .

★ ◆ THEOREM 2.9

Assume that v_1 and v_2 are harmonic conjugates of u in a domain D . Then $v_1 - v_2$ is constant in D .

Proof. Assume that v_1 and v_2 are harmonic conjugates of u in a domain D . Therefore $(v_1)_y = u_x = (v_2)_y$ and $(v_1)_x = -u_y = (v_2)_x$, by the definition of harmonic conjugate. Then $\nabla(v_1 - v_2) =$

0 in D , by direct computation. Then $v_1 - v_2$ is constant in D . Hence the two conjugates differ by a constant. ■

★ ♦ THEOREM 2.10

Assume that $f = u + iv$ is analytic in a domain D and $z_0 \in D$ and $f'(z_0) \neq 0$. Then the level curves $u(x, y) = c_1$ and $v(x, y) = c_2$ through z_0 intersect orthogonally at z_0 .

Proof. Assume that $f = u + iv$ is analytic in a domain D and $z_0 \in D$ and $f'(z_0) \neq 0$. Then $\nabla u \cdot \nabla v = u_x v_x + u_y v_y$, by direct computation. Then $\nabla u \cdot \nabla v = u_x(-u_y) + u_y u_x = 0$. Then $\nabla u(z_0) \neq 0$ and $\nabla v(z_0) \neq 0$, and both level curves are smooth at z_0 with normal vectors $\nabla u(z_0)$ and $\nabla v(z_0)$. Then the normal vectors are orthogonal at z_0 . Hence the level curves intersect orthogonally at z_0 . ■

★ ♦ THEOREM 2.11

Assume that u is harmonic in a domain D and $z_0 \in D$. Then $u(z_0) = \frac{1}{2\pi} \int_0^{2\pi} u(z_0 + re^{i\theta}) d\theta$ for all sufficiently small $r > 0$.